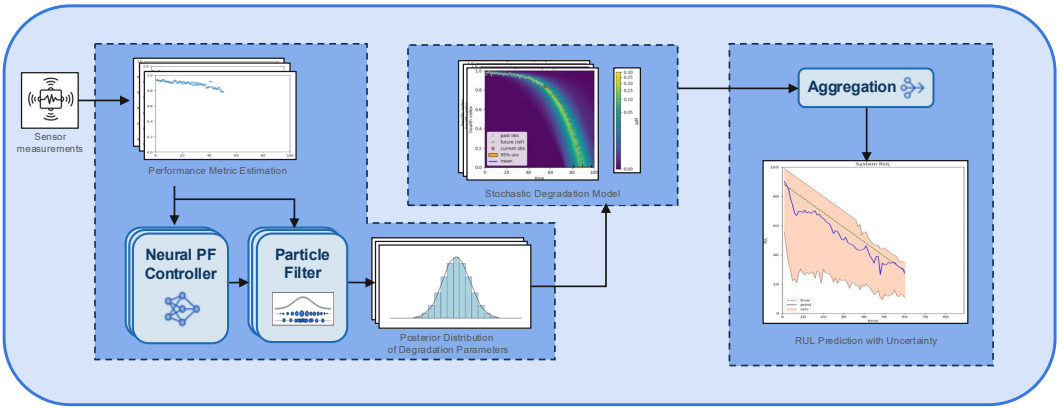
Performance-Based Probabilistic RUL Prediction via Particle Filtering with Neural-Guided Inference.

Estimate System Health → **Predict Degradation** → **Infer RUL Distribution**



Validated Using Multiple N-CMAPSS Benchmark Datasets

- Competitive Accuracy
- Interpretable Predictions
- Calibrated Uncertainty

Separating Estimation and Prediction Improves Interpretable and Uncertainty-Aware RUL Prognostics